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From Script to Tool: Building Robust, Meaningful Tools with PowerShell

Matt McElreath

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April 13-16, 2026

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About Me

- Solution Architect @ FM
- Focus on Infrastructure Atuomation (With PowerShell of course)
- Long time listener, first time presenter

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When do I…?

- "I've learned the PowerShell syntax, gone to the trainings, read the books. How do I use it to improve my work?"
- "When do I use the command line versus a script versus a function versus a Module?"

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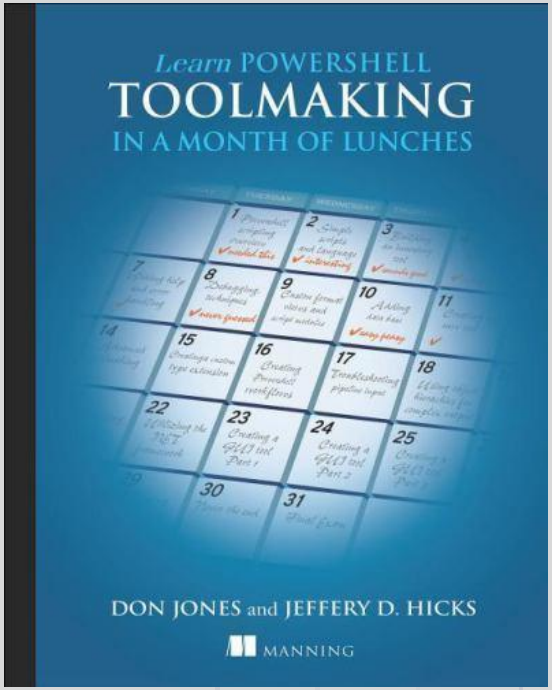

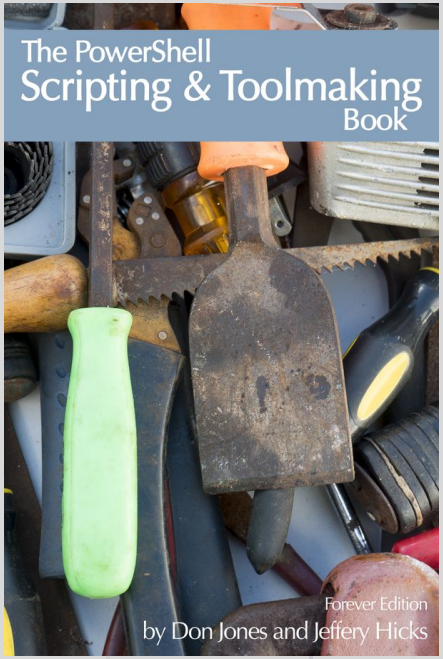

https://www.manning.com/books/learn-powershell-toolmaking-in-a-month-of-lunches
https://leanpub.com/powershell-scripting-toolmaking

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What is Toolmaking In PowerShell

"We borrowed the term *toolmaking* from the tool and die industry... In that industry, there are machinists who produce a variety of different parts and products. To do so, they use tools – drill bits, dies and so forth– which are manufactured by tool and die makers. Both audiences–machinists and the tool and die makers–utilize many of the same skills and equipment, but they do so for different reasons."

- Learn PowerShell Toolmaking in a Month of Lunches
  - Don Jones
  - Jeffery D. Hicks

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What is Toolmaking In PowerShell

- Idea was based off of the tool and die industry
  - Machinists - Produce Parts and Products
  - Tool and Die Makers - Manufacture tools– drill bits, dies and other tools used by Machinists to produce products. A means to an end.
- Most PowerShell users will be using “tools“ developed by others to produce an outcome (command line, combined in scripts, etc…)
- Toolmakers goal create reusable, redistributable tools that can be used to complete tasks

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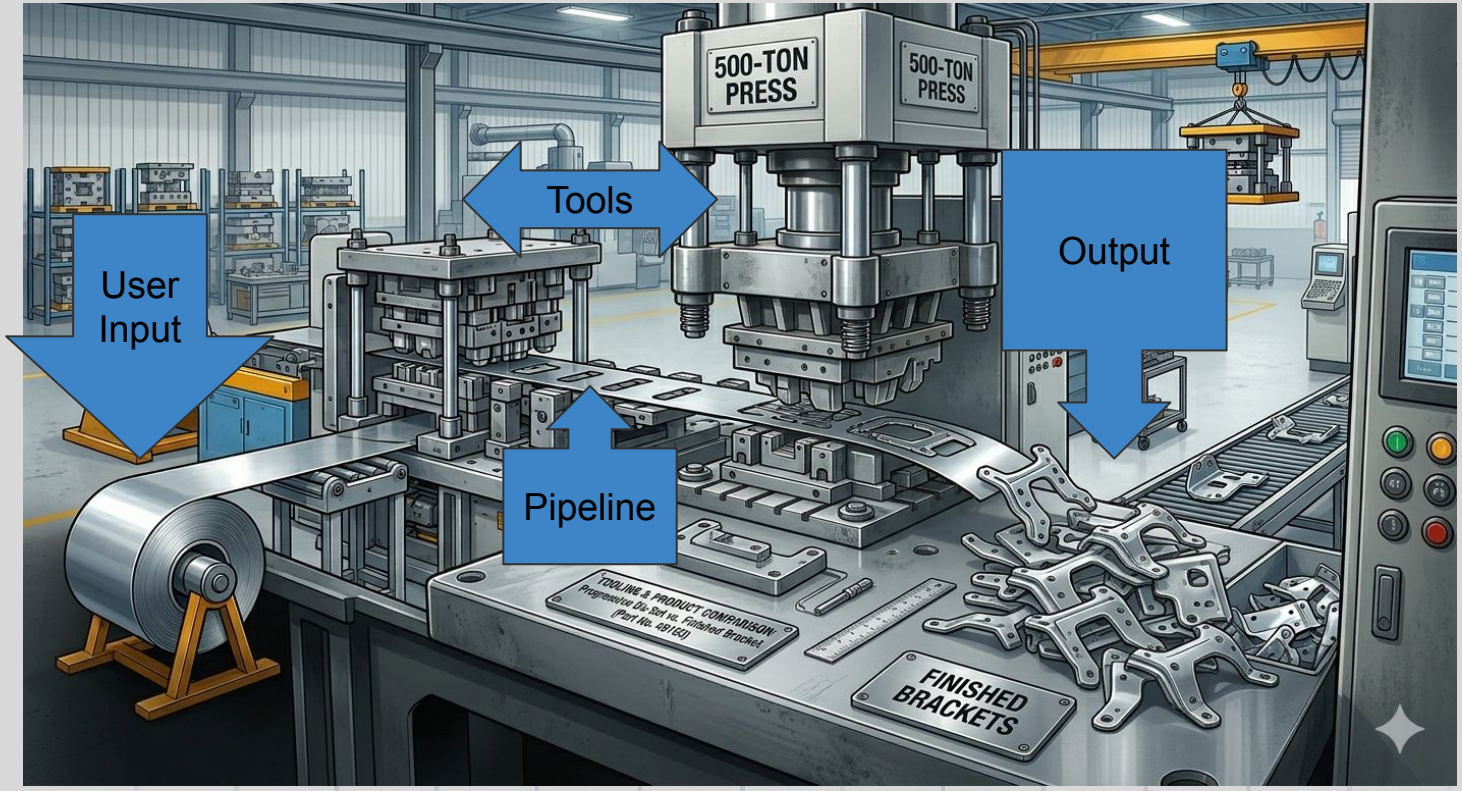
500-TON PRESS
500-TON PRESS
Tools
User Input
Output
Pipeline
TOOLING & PRODUCT COMPARISON
Progressive Die Set vs. Finished Bracket
(Part No. AB1-5X)
FINISHED BRACKETS

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The Toolbox

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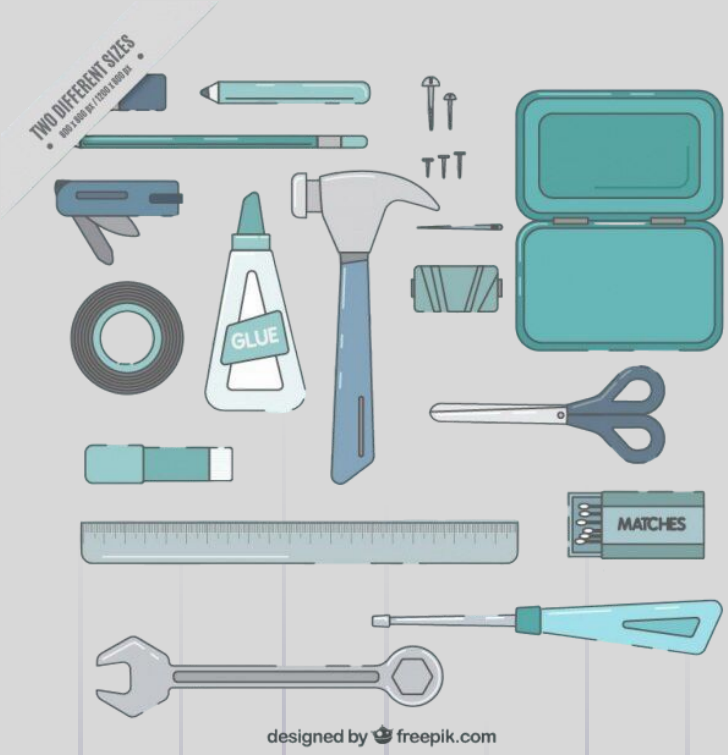

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The Toolbox

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Understand Your Audience

- Who will be using your tool(s)
  - Technical ability
  - PowerShell knowledge/experience
- Different tools for different user types
  - How to accommodate for different user types

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The Toolmaking Journey

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- Hypothesis for Success
- Command Line
- Scripts
- Functions
- Modules

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Starting at the Command Line

- Find the commands/tools for what you're trying to do
  - Ex. Get-ADUser, New-ADUser, Set-ADUser
- Understand their usage
  - What are the parameters
  - How do they interact on the pipeline
  - What can go wrong
    - Easier to troubleshoot from the command line
- What other commands can you use to interact with data
  - ConvertTo-CSV, ConvertFrom-JSON, etc…

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Demo

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Put your commands in a .ps1 file

- Great for repeatability
- OK for redistribution
- Can be Parameterized for flexibility
- Specify Parameter types for consistency
- Parameter Validation!!!!
- Comments, Comments, Comments

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Demo

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Putting your commands in functions

- Paramaterized, reusable code within your script
  - Avoid duplicating code
  - Improve Readabilty
- Compared to scripts, multiple functions don't need separate files. Can be bundled into a single file
- Comments, Comments, Comments

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Demo

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PowerShell Modules

- Don't be scared
  - Learn the structure of a PowerShell Module
  - The rest is just the same as the last couple slides
- **Use [CmdletBinding()]**
- Functions can be structured into folders containing multiple .ps1 files
  - Ex. Public and Private functions
- Help documentation
- Distribution is easy
  - PowerShell Gallery
  - Private Gallery
- Comments, Comments, Comments

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Demo

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C.R.U.D

- The basic operations of persistent storage
  - Create - New-ADUser
  - Read - Get-ADUser
  - Update - Set-ADUser
  - Delete - Remove-ADUser

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Outputs

- Basic Text

- JSON
- Hash Table

- PSObjects

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Error Handling

- How does your tool handle Terminating and Non-Terminating errors
- Use Try/Catch
- Surfacing meaningful error messages

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Don't Cling

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Resources

- PowerShell.Org YouTube Channel
  https://www.youtube.com/@PowershellOrg

- Basic Toolmaking - The road to extensibility by Evgenij Smirnov
  https://www.youtube.com/watch?v=atWtV9UUjLI

- The Human Side of PowerShell Scripting by Jeff Hicks
  https://www.youtube.com/watch?v=4xVDorpEhm8

- https://www.manning.com/books/learn-powershell-toolmaking-in-a-month-of-lunches

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THANK YOU

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